



# Python Programming Fundamentals

## Why Collections?

Programming Fundamentals Team

SETU

2026-06-17



# Outline

1. The Problem
2. Python Lists as Arrays
3. Why This Matters
4. Today's Goal



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# 1. The Problem

So far, we can store one value at a time.

```
student1 = "Ava"  
student2 = "Ben"  
student3 = "Cara"
```



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So far, we can store one value at a time.

```
student1 = "Ava"  
student2 = "Ben"  
student3 = "Cara"
```

This works for three students, but not for thirty.

**Collections** let us store many related values under one name.



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## 2. Python Lists as Arrays

In many languages, students first meet **arrays**.

In Python, the closest beginner-friendly equivalent is a **list**.

```
students = ["Ava", "Ben", "Cara"]  
grades = [72, 81, 65, 90]
```



## 2. Python Lists as Arrays

In many languages, students first meet **arrays**.

In Python, the closest beginner-friendly equivalent is a **list**.

```
students = ["Ava", "Ben", "Cara"]  
grades = [72, 81, 65, 90]
```

A list is:

- ordered
- changeable
- indexed from 0
- able to grow and shrink



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## 3. Why This Matters

Collections are used everywhere:

- grades in a class
- items in a shopping basket
- sensor readings
- names in a booking system
- records loaded from a file



### 3. Why This Matters

Collections are used everywhere:

- grades in a class
- items in a shopping basket
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They also form the basis of later topics:

- files and JSON
- CRUD programs
- classes and objects
- data processing



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## 4. Today's Goal

By the end of this session, you should be able to:

- create a list
- access items by position
- change list contents
- use strings as sequences
- slice lists and strings
- process lists using loops



# Thanks for Watching - Any questions?

